20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?

Jan Moritz Beyn

2026-08-07

Abstract

Here I write normal text.

Introduction

Theory/Literature Review

Hypotheses

Analysis

Data Cleaning

To be able to analyze the data from the different elections and compare the outcomes. Each dataset has to be cleaned. For some cantons e.g. Valais and Schwyz the data is very scattered, therefore for these multiple datasets per election have to be cleaned and merged.

The procedure for the data cleaning was the following: For each canton in alphabetically order, for each election the data was cleaned.

To ensure the compatibility of the different datasets, the column names were standardized. The column names were renamed for each dataset to the following: “total_votes”, “svp_votes”, “sp_votes”, “cvp_votes”, “fdp_votes”, “glp_votes”, “bdp_votes”, “green_votes”, “other_votes”. It might be that some of the datasets do not contain all of these columns, since the party landscapes varies geographically and over time.

```
# Import and inspect the raw Aargau 2009 dataset
ag_09_raw <- read_excel(
  here("data", "datasets", "aargau", "AG_09_kantonsratswahlen.xlsx"),
  sheet = "Data"
```

```
)

glimpse(ag_09_raw)
```

```
Rows: 241
Columns: 21
$ bfsNr        <dbl> 19, 1901, 1902, 1903, 1904, 1905, 1906, 1907, 1908, 1909,…
$ year         <dbl> 2009, 2009, 2009, 2009, 2009, 2009, 2009, 2009, 2009, 200…
$ month        <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, …
$ day          <dbl> 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, …
$ locationName <chr> "Kanton Aargau", "Bezirk Aarau", "Bezirk Baden", "Bezirk …
$ locationType <chr> "CANTON", "DISTRICT", "DISTRICT", "DISTRICT", "DISTRICT",…
$ TOTAL        <dbl> 1830693, 240252, 705906, 205249, 111613, 63448, 44191, 12…
$ SVP          <dbl> 556964, 72796, 189366, 66189, 36228, 23440, 15559, 45138,…
$ SP           <dbl> 291085, 47330, 106510, 27604, 17504, 8360, 5717, 21596, 5…
$ CVP          <dbl> 275911, 21319, 131935, 39901, 10580, 2554, 11438, 8054, 1…
$ FDP          <dbl> 264828, 43245, 99220, 25582, 21661, 10744, 4886, 17379, 4…
$ GRUENE       <dbl> 164610, 23528, 64744, 16035, 12601, 5952, 3607, 7866, 306…
$ EVP          <chr> "83096.0", "14365.0", "28138.0", "3522.0", "6931.0", "558…
$ SD           <chr> "21239.0", "2463.0", "6757.0", "1950.0", "...", "2719.0",…
$ EDU          <chr> "30609.0", "7227.0", "6544.0", "1603.0", "1574.0", "2649.…
$ BDP          <chr> "71489.0", "...", "41512.0", "11629.0", "...", "...", "..…
$ UNS          <chr> "1196.0", "1196.0", "...", "...", "...", "...", "...", ".…
$ GLP          <chr> "68113.0", "6783.0", "30882.0", "10055.0", "4534.0", "144…
$ NPAK         <chr> "76.0", "...", "...", "...", "...", "...", "...", "...", …
$ L_PLUS       <chr> "1179.0", "...", "...", "1179.0", "...", "...", "...", ".…
$ FAAG         <chr> "298.0", "...", "298.0", "...", "...", "...", "...", "...…
```

```
print(ag_09_raw, n = 20)
```

```
# A tibble: 241 × 21
   bfsNr  year month   day locationName locationType  TOTAL    SVP     SP    CVP
   <dbl> <dbl> <dbl> <dbl> <chr>        <chr>         <dbl>  <dbl>  <dbl>  <dbl>
 1    19  2009     3     8 Kanton Aarg… CANTON       1.83e6 556964 291085 275911
 2  1901  2009     3     8 Bezirk Aarau DISTRICT     2.40e5  72796  47330  21319
 3  1902  2009     3     8 Bezirk Baden DISTRICT     7.06e5 189366 106510 131935
 4  1903  2009     3     8 Bezirk Brem… DISTRICT     2.05e5  66189  27604  39901
 5  1904  2009     3     8 Bezirk Brugg DISTRICT     1.12e5  36228  17504  10580
 6  1905  2009     3     8 Bezirk Kulm  DISTRICT     6.34e4  23440   8360   2554
 7  1906  2009     3     8 Bezirk Lauf… DISTRICT     4.42e4  15559   5717  11438
 8  1907  2009     3     8 Bezirk Lenz… DISTRICT     1.25e5  45138  21596   8054
```

```
 9  1908  2009     3     8 Bezirk Muri  DISTRICT     4.27e4  13893   5225  12752
10  1909  2009     3     8 Bezirk Rhei… DISTRICT     6.70e4  20511  12840  10660
11  1910  2009     3     8 Bezirk Zofi… DISTRICT     1.75e5  54312  32663  11971
12  1911  2009     3     8 Bezirk Zurz… DISTRICT     5.01e4  19532   5736  14747
13  4001  2009     3     8 Aarau        TOWNSHIP     6.81e4  13936  16727   6550
14  4271  2009     3     8 Aarburg      TOWNSHIP     1.30e4   3189   3484   1419
15  4221  2009     3     8 Abtwil       TOWNSHIP     1.20e3    432     53    521
16  4191  2009     3     8 Ammerswil    TOWNSHIP     1.70e3    716    355     60
17  4222  2009     3     8 Aristau      TOWNSHIP     1.52e3    384    214    462
18  4061  2009     3     8 Arni (AG)    TOWNSHIP     3.85e3   1194    340    705
19  4272  2009     3     8 Attelwil     TOWNSHIP     1.34e3    667     63     77
20  4091  2009     3     8 Auenstein    TOWNSHIP     3.82e3    876    532    175
# ℹ 221 more rows
# ℹ 11 more variables: FDP <dbl>, GRUENE <dbl>, EVP <chr>, SD <chr>, EDU <chr>,
#   BDP <chr>, UNS <chr>, GLP <chr>, NPAK <chr>, L_PLUS <chr>, FAAG <chr>
```

```
names(ag_09_raw)
```

```
 [1] "bfsNr"        "year"         "month"        "day"          "locationName"
 [6] "locationType" "TOTAL"        "SVP"          "SP"           "CVP"
[11] "FDP"          "GRUENE"       "EVP"          "SD"           "EDU"
[16] "BDP"          "UNS"          "GLP"          "NPAK"         "L_PLUS"
[21] "FAAG"
```

```
# Create and clean a new dataset in memory
ag_09_clean <- ag_09_raw |>
  clean_names() |>
  rename( # Rename columns to standardized names
    total_votes = total,
    svp_votes = svp,
    sp_votes = sp,
    cvp_votes = cvp,
    fdp_votes = fdp,
    glp_votes = glp,
    bdp_votes = bdp,
    gruene_votes = gruene,
  ) |>
  select(-bfs_nr, -month, -day) |> # delete unnecessary columns
  slice(-(13:241)) # drop not needed municipality level columns
```

```r
# Inspect the cleaned data
glimpse(ag_09_clean)
```

```
Rows: 12
Columns: 18
$ year          <dbl> 2009, 2009, 2009, 2009, 2009, 2009, 2009, 2009, 2009, 20…
$ location_name <chr> "Kanton Aargau", "Bezirk Aarau", "Bezirk Baden", "Bezirk…
$ location_type <chr> "CANTON", "DISTRICT", "DISTRICT", "DISTRICT", "DISTRICT"…
$ total_votes   <dbl> 1830693, 240252, 705906, 205249, 111613, 63448, 44191, 1…
$ svp_votes     <dbl> 556964, 72796, 189366, 66189, 36228, 23440, 15559, 45138…
$ sp_votes      <dbl> 291085, 47330, 106510, 27604, 17504, 8360, 5717, 21596, …
$ cvp_votes     <dbl> 275911, 21319, 131935, 39901, 10580, 2554, 11438, 8054, …
$ fdp_votes     <dbl> 264828, 43245, 99220, 25582, 21661, 10744, 4886, 17379, …
$ gruene_votes  <dbl> 164610, 23528, 64744, 16035, 12601, 5952, 3607, 7866, 30…
$ evp           <chr> "83096.0", "14365.0", "28138.0", "3522.0", "6931.0", "55…
$ sd            <chr> "21239.0", "2463.0", "6757.0", "1950.0", "...", "2719.0"…
$ edu           <chr> "30609.0", "7227.0", "6544.0", "1603.0", "1574.0", "2649…
$ bdp_votes     <chr> "71489.0", "...", "41512.0", "11629.0", "...", "...", ".…
$ uns           <chr> "1196.0", "1196.0", "...", "...", "...", "...", "...", "…
$ glp_votes     <chr> "68113.0", "6783.0", "30882.0", "10055.0", "4534.0", "14…
$ npak          <chr> "76.0", "...", "...", "...", "...", "...", "...", "...",…
$ l_plus        <chr> "1179.0", "...", "...", "1179.0", "...", "...", "...", "…
$ faag          <chr> "298.0", "...", "298.0", "...", "...", "...", "...", "..…
```

```r
print(ag_09_clean, n = Inf)
```

```
# A tibble: 12 × 18
    year location_name    location_type total_votes svp_votes sp_votes cvp_votes
   <dbl> <chr>            <chr>               <dbl>     <dbl>    <dbl>     <dbl>
 1  2009 Kanton Aargau    CANTON            1830693    556964   291085    275911
 2  2009 Bezirk Aarau     DISTRICT           240252     72796    47330     21319
 3  2009 Bezirk Baden     DISTRICT           705906    189366   106510    131935
 4  2009 Bezirk Bremgart… DISTRICT           205249     66189    27604     39901
 5  2009 Bezirk Brugg     DISTRICT           111613     36228    17504     10580
 6  2009 Bezirk Kulm      DISTRICT            63448     23440     8360      2554
 7  2009 Bezirk Laufenbu… DISTRICT            44191     15559     5717     11438
 8  2009 Bezirk Lenzburg  DISTRICT           124690     45138    21596      8054
 9  2009 Bezirk Muri      DISTRICT            42712     13893     5225     12752
10  2009 Bezirk Rheinfel… DISTRICT            67012     20511    12840     10660
11  2009 Bezirk Zofingen  DISTRICT           175478     54312    32663     11971
12  2009 Bezirk Zurzach   DISTRICT            50142     19532     5736     14747
```

```
# ℹ 11 more variables: fdp_votes <dbl>, gruene_votes <dbl>, evp <chr>,
#   sd <chr>, edu <chr>, bdp_votes <chr>, uns <chr>, glp_votes <chr>,
#   npak <chr>, l_plus <chr>, faag <chr>
```

The canton of Aargau, used prior to the Double Pukelsheim system the Hagenbach-Bischoff method. While the HB System was in use, some parties formed alliances (Staatskanzlei des Kantons Aargau, 2005, S. 8). For the main counterfactual analysis, the seats were calculated on base of that these alliances would not have been formed. For a Robustness check, the seats were also calculated on base of the actual votes, which includes the alliances. The results of both calculations are compared to the actual seat distribution.

Results

Interpretation of the Results

Discussion

Conclusion

Bibliography

Staatskanzlei des Kantons Aargau. (2005, 7. März). Bezirksprotokolle über die Erneuerungswahl des Grossen Rats für die Amtsperiode 2005/2009: Wahltag: 27. Februar 2005 (Amtsblattbeilage 2005-1). https://doczz.net/doc/5954970/amtsblattbeilage_vom_07.03.2005

Appendix

Appendix 1: Explanation Data Cleaning Process

Appendix 2: Code for Data Cleaning

Declaration of AI-Assistance